Chapter 3

Reproducibility, Degradation, and the Fabrication-Record Methodology

3.1 The Reproducibility Problem

Chapter 2 established the memristive response of the formulation comprising *Super Yellow*, *Hybrane*, and LiOTf across assay-specific junctions and reported two-week ambient shelf stability. In the subsequent fabrication campaign, memristive hysteresis became progressively harder to obtain. Successive batches of nominally identical devices shifted from the low-current, high-area, multi-level response of the proof-of-concept platform toward high conductivity, a narrow switching window, and no usable potentiation. This chapter analyses that loss of reproducibility and the subsequent change of host before chapter 4 examines a replicated composition grid.

Individual-device shelf stability and campaign-scale reproducibility measure different processes. The proof-of-concept shelf record retained its

characteristics over days to weeks, as did other finished *Hybrane* devices. The fabrication line nevertheless changed between batches over the following year. This places the degradation in the supply or fabrication process over calendar time, while the finished devices remained stable over the measured shelf interval; section 3.3 quantifies the distinction.

The fabrication and characterisation notes identify devices NM_v026–031 (2021-04-22) as a change point, recorded at the time as "the first devices that changed electrical behaviour". The contemporaneous hypotheses remained open ("could be evaporation rate ... the metal used for evaporation, and possible chemical degradation of substances") and were still unresolved in a note dated 2022-01-31, after nine months of investigation. The analysis uses this documented change point as a pre/post boundary and does not assume a smooth trend. It tests the candidate causes, quantifies the degradation under four controls, and evaluates the change to PEO.

3.2 Eliminating Candidate Causes

The team tested the candidate causes one at a time in batches with the other variables held fixed. Table 3.1 lists each cause, the batches that tested it, and the resulting verdict. This process also motivated the fabrication-record library used later in the thesis, since each test required the process variables to be recorded per device.

Device NM_v067 (2021-10-21) records that a second experimenter, working independently, "has the same problems with devices". This cross-person result excludes individual technique as the cause and indicates a shared material or process source. The same note proposed the ITO substrate, but the substrate study did not support that explanation. The *Super Yellow* semiconductor was also tested because an old bottle was

Table 3.1: Methods-of-elimination summary for the degradation investigation. Candidate causes were tested in deliberate batches with variables recorded in the fabrication record. After the exclusions, the hyperbranched *Hybrane* host stock remains the candidate examined quantitatively in section 3.3.

Candidate cause	**Crossed as (batches)**	**Verdict**
Operator / technique	cross-person check (v067): "Lorenzo has the same problems with devices"	**Eliminated.** The same result occurred with a second experimenter, excluding one operator's technique.
Illumination (light vs dark)	v015–019, vt001–006, v041–044, v053–056	Excluded (no systematic recovery of the window).
Storage / measurement atmosphere (glovebox / air / vacuum)	v015–019, v088–099	Excluded.
Baby-chamber transport (full 2×2 on Au)	v083–087, v038–040	Excluded.
Substrate (ITO) lot / colour	v079–082; substrate-difference study	Not borne out (a one-off ITO guess by a colleague did not survive the substrate comparison).
Solvent (cyclohexanone) lot	v079, v091	Excluded.
Evaporation rate / silver holder	v045–048, v106–113	Recovery *attempted* by reverting to the "year-ago" holder; not reproducibly restored.
Annealing regime	late hot-anneal batches	*Partial* window recovery (section 3.3); consistent with the chemical hypothesis but insufficient as a process correction.
Super Yellow stock (old vs new)	v045–048 ("more viscous … maybe bonds degradation in the old one")	Eliminated as primary (old vs new *Super Yellow* gave no significant change, and fresh *Super Yellow* continued to be purchased while the window loss persisted).
***Hybrane* host stock**	residual after the above	**Remaining candidate**, examined quantitatively and mechanistically below.

noticeably "more viscous … maybe because of bonds degradation in the old one". The old-versus-new *Super Yellow* comparison produced no significant change, and devices retained the reduced window while fresh *Super Yellow* was used. After illumination, atmosphere, transport, substrate, solvent, evaporation, and semiconductor were tested and excluded, the hyperbranched *Hybrane* host stock remained the candidate consumable that had aged between batches. The remainder of the chapter tests that attribution quantitatively and proposes a candidate chemistry.

3.3 The Quantitative Degradation Signal

The device archive provides a quantitative test of the degradation recorded in the laboratory notes. The analysis controls for the sampling and protocol changes within the campaign and is reproduced in full by `scripts/bridge_hybrane_peo_reproducibility.py`; it mirrors the slicing of the group's interactive feature-explorer tool.

Four controls, each shown to matter. The analysis rests on four controls, each verified to change the conclusion if dropped: *per-device weighting* (the pixels-per-device count falls with date, Spearman $\rho = -0.80$, so every feature is reduced to one value per device before any test); a *standard-protocol corpus* ($n \approx 69$ devices, lead mass ratios, excluding the invasive recovery experiments); *sweep-amplitude matching* (the window features are read in the $\sim$3 V stratum where they are expressed, the conductivity features in a tightly matched $\sim$1.2 V band); and a *pre/post-inflection step test* fixed at the documented NM_v026 inflection, with every conductivity trend reported with and without the deliberately-reverted recovery batches. The four controls are stated in full, with the supporting correlations, in Appendix B.

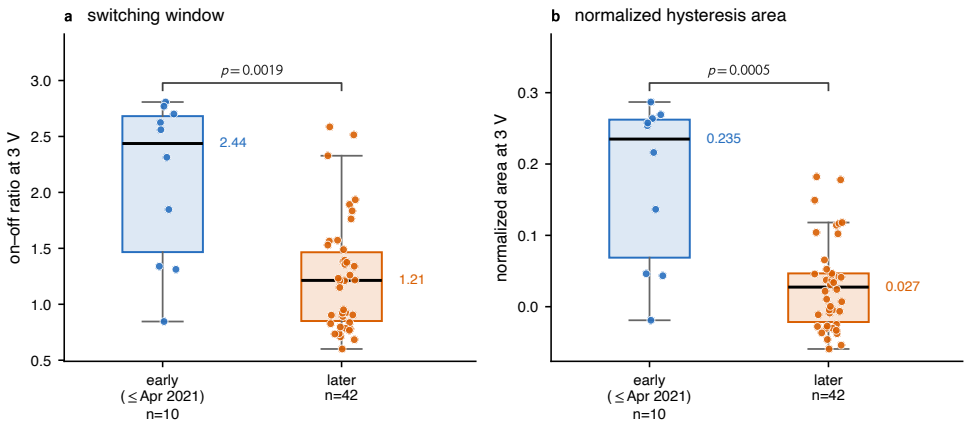


Figure 3.1: Switching-window reduction across the campaign on the standard *Super Yellow*/*Hybrane*/LiOTf/Ag corpus, read at the ∼3 V stratum where the window is expressed, one value per device. (a) On–off ratio and (b) normalized hysteresis area, both for devices fabricated before the NM_v026 change point (≤Apr 2021) versus later. Boxes are quartiles with the median annotated beside each box; points are individual devices; the bracket gives the two-sided Mann–Whitney p. The on–off ratio decreases by roughly a factor of two and the area by an order of magnitude. Reproduced by `scripts/bridge_figures.py` (F1).

Principal result: the switching window decreases at ∼3 V. On the standard corpus, per device, the switching window measured in the ∼3 V stratum decreases across the change point (fig. 3.1, table 3.2). The median on–off ratio falls from 2.44 (early, $n = 10$) to 1.21 (later, $n = 42$; Mann–Whitney $p = 0.0019$); the median normalized hysteresis area falls from 0.235 to 0.027, an order of magnitude ($p = 0.0005$); and the potentiation, measured as the per-cent change in the maximum-voltage current over a sweep, changes from +17.9 % to −2.1 % ($p = 0.013$). The on–off ratio, hysteresis area, and potentiation thus identify the same loss of switching function. The magnitude agrees with the team's recollection of an early on–off near 2.5 falling to the 1–1.7 range later, now tied to the documented change point.

Table 3.2: The degradation as a multi-feature signal on the standard corpus, per device. *Window reduction* is read at ∼3 V as an early-vs-later step at the NM_v026 change point (Mann–Whitney). *Ohmic drift* is read at matched ∼1.2 V as a per-device Spearman correlation with fabrication date, reported with all dates and with the recovery tail (≥2022-03) dropped; the correlation increases when the reverted batches are removed, as expected if they are partial recoveries. Health flags are included and discussed in the text.

Window reduction @ ∼3 V (early ≤Apr 2021, $n = 10$ vs later, $n = 42$)			
Feature	early	later	Mann–Whitney
on–off ratio	2.44	1.21	$p = 0.0019$
normalized area	0.235	0.027	$p = 0.0005$
% change in max-V current	+17.9 %	−2.1 %	$p = 0.013$
Ohmic drift @ matched ∼1.2 V (per-device Spearman vs date, $n = 64$)			
Feature	all dates	drop recovery tail	
current at max V	$\rho = +0.45$ ($p = 0.0002$)	$\rho = +0.55$ ($p < 10^{-4}$)	
hysteresis area (V·μA)	$\rho = +0.57$ ($p < 10^{-4}$)	$\rho = +0.69$ ($p < 10^{-4}$)	
current difference at on–off	$\rho = +0.51$ ($p < 10^{-4}$)	$\rho = +0.63$ ($p < 10^{-4}$)	
Health flags (per-device, ∼1.2 V, $n = 64$)			
`is_broken` *decreases* ($\rho = -0.40$, $p = 0.001$), a handling metric, *not* degradation; `is_saturated` (fails to potentiate) *rises* ($\rho = +0.27$, $p = 0.03$).			

Mechanism: ohmic drift at matched ∼1.2 V. Conductivity provides a complementary signal, read in the tightly matched ∼1.2 V band so that the sweep-amplitude confound is held fixed. The absolute current rises with fabrication date: per device, the current at maximum voltage correlates with date at $\rho = +0.45$ ($p = 0.0002$), the raw hysteresis area at $\rho = +0.57$, and the current difference at the on–off point at $\rho = +0.51$ (all $n = 64$; fig. 3.2a, table 3.2). These correlations increase when the deliberately reverted recovery batches (≥2022-03) are dropped (to $\rho = +0.55$, $+0.69$, and $+0.63$ respectively), as expected if those late batches are partial recoveries. Because the amplitude is held within a narrow band, the trend indicates more ohmic conduction independently of sweep range. The reduction in relative hysteresis and the increase in

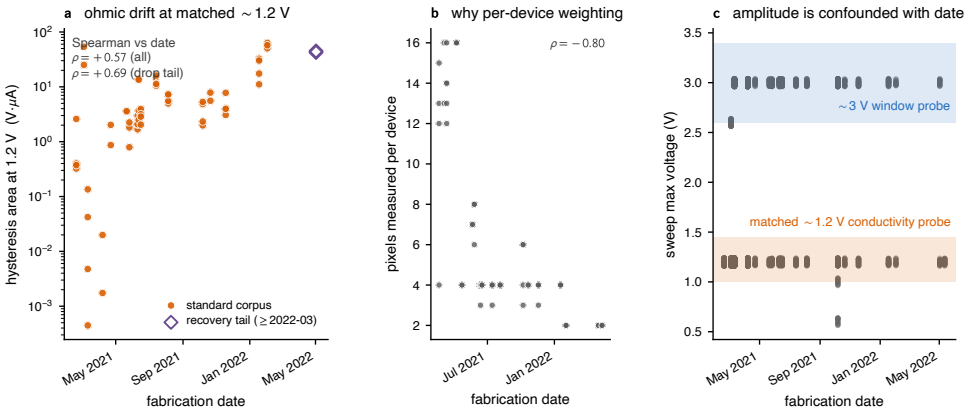


Figure 3.2: Mechanism and methodology. (a) Per-device hysteresis area read at the matched ∼1.2 V band rises with fabrication date (Spearman $\rho = +0.57$ over all dates, strengthening to $+0.69$ when the recovery-tail batches, open diamonds, ≥2022-03, are dropped; the statistics use all $n = 64$ devices, while the log axis cannot display the few non-positive areas). (b) The number of pixels measured per device falls over the campaign ($\rho = -0.80$), which is why every statistic is computed per device rather than per pixel. (c) Sweep amplitude is itself confounded with date: the matched ∼1.2 V conductivity probe and the ∼3 V window probe (shaded) are the two strata used throughout, never pooled. Reproduced by `scripts/bridge_figures.py` (F2).

absolute current are two measurements of the same degradation. The magnitudes are reported as correlations and medians because absolute current depends on sweep amplitude.

Interpretation of the health flags. The fraction of broken devices would mischaracterize the degradation. The curve-level `is_broken` flag (erratic or open-circuit behaviour caused by contact or handling problems) decreases over the campaign ($\rho = -0.40$, $p = 0.001$) as handling and contacting improved. The `is_saturated` flag (a curve whose maximum current fails to exceed the previous one, indicating loss of potentiation) rises with date ($\rho = +0.27$, $p = 0.03$). The two flags are reported separately because they measure different processes.

3.3.1 A Candidate Chemistry: Moisture-Driven Hydrolytic Aging of the *Hybrane* Stock

The available record supports attribution to the stock but cannot establish a chemical mechanism. No electrochemical impedance spectra were recorded on a *Hybrane* device; no gel-permeation chromatography, nuclear-magnetic-resonance, or Karl-Fischer analysis was performed on the reagent stock; and the aged stock no longer exists in its degraded state. The proposed moisture-driven hydrolysis mechanism therefore remains a hypothesis, accompanied by experiments that could test it.

The hypothesis. *Hybrane* is a hyperbranched polyester-amide [1, 2]: ester- and amide-rich, and strongly hygroscopic. The leading hypothesis is that, over roughly a year in the bottle, the stock takes up atmospheric water and slowly hydrolyses (chain scission), and that this process could account for the measured electrical signatures without requiring a separate finished-device ageing mechanism.

- **Ohmic drift** (current $\uparrow$ at matched 1.2 V): absorbed water together with polar scission products raises the baseline ionic and leakage conductivity and plasticises the matrix.

- **Window reduction** (on–off $\rightarrow\sim$1.2, area $\rightarrow\sim$0.03, $\%\Delta I \rightarrow -2\,\%$): in the metastable-polarisation model of chapter 2, memristance requires a matrix that retains a field-displaced ionic profile. A plasticised, lower-molecular-weight, water-laden host would relax more rapidly, reducing the loop area and potentiation across successive sweeps.

- *(secondary)* acidic hydrolysis products could partially dope the PPV-type *Super Yellow*, adding a fixed, irreversible conductance, a

second route to the “high-conductivity, undesirable” state recorded in the notes.

Internal support. Several observations are consistent with the hypothesis but do not establish it. Among late devices, annealing at $\geq$150 °C increases the median on–off ratio at $\sim$3 V to 1.95 ($n = 6$), compared with 1.22 for standard 75 °C late devices ($n = 44$; one-sided Mann–Whitney $p = 0.024$), near the early-standard reference of 1.74 (fig. 3.3). Removal of absorbed water and film densification could account for this recovery. The test remains limited by six elevated-temperature devices, co-varying process changes, and measurement-pin issues in two devices. The combination of ohmic drift, window reduction, and loss of potentiation is also consistent with a water-plasticised, ion-leaky film. Chemically, the ester and amide linkages make this material class susceptible to hydrolytic chain scission, while its polar, hyperbranched architecture is hygroscopic [1, 2].

Confirming experiments (future work). The hypothesis would be tested directly by Fourier-transform infrared spectroscopy (loss of the ester C=O band, growth of –OH) and gel-permeation chromatography (a molecular-weight drop) on *retained aged versus fresh* stock; by Karl-Fischer titration or thermogravimetry for water content; and by impedance spectroscopy to show that the ionic resistance fell. None were run on the *Hybrane* corpus, so the hypothesis stays open, and may never fully close, because it concerns a consumable that no longer exists in its aged state.

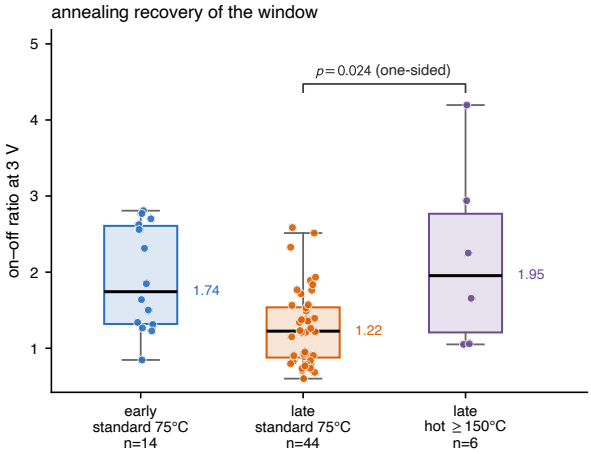


Figure 3.3: Annealing-recovery test of the hydrolysis hypothesis (*Hybrane*/Ag, on–off ratio at ∼3 V, per device). Among late devices, annealing at ≥150 °C increases the median from 1.22 to 1.95, near the early-standard reference of 1.74; one-sided Mann–Whitney $p = 0.024$. Interpretation is limited by $n = 6$ high-temperature devices, co-varying process changes, and pin issues in two devices. Reproduced by `scripts/bridge_figures.py` (F4).

3.4 Change to a PEO Host

The fabrication programme changed the ion-transport host after attempts to recover the *Hybrane* line. Replacing hyperbranched *Hybrane* with linear poly(ethylene oxide) (PEO), the host used in the comparative study of chapter 4, produced a larger switching window and multi-year reproducibility. Laboratory notes first record PEO in use by 2021-12 (NM_v083).

Switching-window comparison. At a matched ∼3 V sweep, comparing valid (non-broken, non-saturated) devices per device, the PEO host gives a larger switching window than contemporaneous *Hybrane* devices: median normalized area 0.26 (*Hybrane*) → 0.42 (PEO), Mann–Whitney $p = 0.018$; median on–off ratio 2.45 → 4.91 (fig. 3.4a,b).

Sustained temporal reproducibility. The PEO/Ag platform retained a usable switching window over roughly two and a half years (2022–2024, $n = 46$ devices): the per-device on–off ratio at $\sim$3 V had a median of 2.61, with yearly medians of 3.3/2.6/1.9 and no significant downward trend over time. The *Hybrane* platform had a median on–off ratio of 1.35 after its stock-associated window reduction (fig. 3.4c). The PEO record shows no corresponding supply-degradation trend over the measured interval.

Meaning of "reproducible". Here, reproducibility refers to the absence of a stock-associated reduction between batches over several years. It does not imply superior within-batch device uniformity: at matched $\sim$3 V, the device-to-device coefficients of variation differ only modestly (PEO $\approx$0.38 vs *Hybrane* $\approx$0.47). The evidence for PEO is its larger switching window and the absence of the supply-degradation trend over the multi-year record. Changing the suspected *Hybrane* reagent removed the trend after *Super Yellow* had been eliminated as its source (section 3.2).

3.5 Methodological Outputs

The investigation required four tools that were not available at the start of the campaign: a fabrication record, a normalized characterization procedure, an automated feature pipeline, and trend-discovery software. The same tools produce the quantitative results in the comparative and temporal-computing chapters. Five outputs are retained from this work.

- **A fabrication-record standard.** The investigation produced a per-device library schema for storage atmosphere, illumination, transport, reagent and solvent lots, operator, evaporation profile, and free-text fabrication and characterisation notes. This record

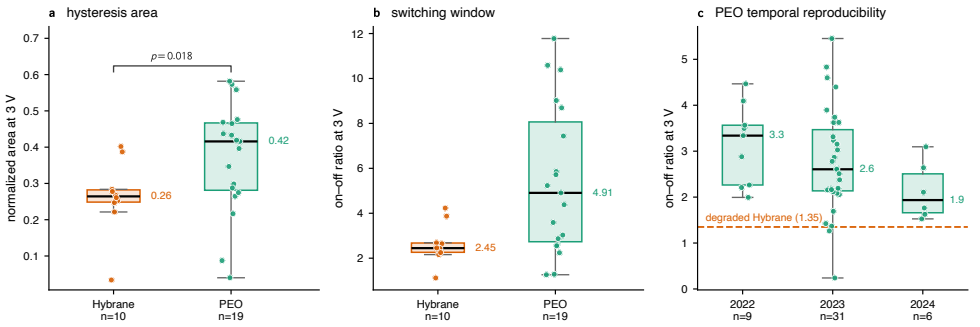


Figure 3.4: Switching-window response after changing to a PEO host (silver electrode, ∼3 V, per device). (a) Normalized hysteresis area and (b) on–off ratio are larger for PEO than for contemporaneous valid *Hybrane* devices ($p = 0.018$ for area). (c) The per-device PEO on–off ratio remains at a usable level (median ≈2.6) across 2022–2024 without a significant downward trend; the dashed line gives the degraded-*Hybrane* level (1.35). Reproduced by `scripts/bridge_figures.py` (F3).

enabled the exclusions in section 3.2 and is retained for every device in the archive.

- **A normalized electrical-characterization procedure.** Section 3.3 shows that sweep amplitude affects apparent conductivity and whether a switching window is expressed. The campaign therefore converged on fixed, documented drive and read protocols that support comparisons between devices and batches.

- **An automated feature pipeline.** The raw instrument traces are reduced to per-device, per-pixel, and per-curve feature tables by a dedicated pipeline; every feature used in this chapter's panel, and in chapter 4 and the temporal-computing chapter, is computed there.

- **Trend-discovery and curation tooling.** An interactive feature explorer identified the reported trends, whose slices are reproduced by the four controls of section 3.3; a curation tool produces the screened device set used in the analysis.

- **Figure generation.** A thesis-figure pipeline renders the quantitative figures from the same feature tables, so that every plotted number is traceable to the archive.

The comparative study of chapter 4 and the data-driven temporal-computing analysis use this same apparatus.

3.6 Reconciliation of Stability and Reproducibility

The shelf-stability result of chapter 2 and the campaign-scale loss of reproducibility concern the two different measurements defined in section 3.1.

- The published proof-of-concept shelf record was stable on the measured ∼two-week ambient scale [3]. A separate multi-day study likewise shows that finished *Hybrane* devices retain their characteristics over weeks to months. These measurements associate the campaign-scale change with the supply or fabrication process, while finished devices remained stable over the measured intervals.
- The *batch-scale reproducibility of the Hybrane platform over the fabrication campaign* is a different metric, and it declined over the following year as quantified in section 3.3.

Together, the records describe short-interval shelf stability within the proof-of-concept platform and a fabrication line over a year. The exact specimen roster of the published shelf record is not resolved in the recovered archive, and the campaign-scale reproducibility axis was not measured in the proof-of-concept study.

After the switch to PEO, replicated measurements of the volatile dynamics became possible. Chapter 4 uses the *Super Yellow*/PEO/LiOTf system

on silver as a composition grid, measured with the fabrication-record standard, normalized protocols, and feature pipeline developed during the reproducibility investigation. Because every substrate now carries its fabrication record, the substrate-to-substrate spread of each composition can be measured, and Chapter 5 uses it as the tolerance of a composition-defined element.

References

1. Froehling, P. Development of DSM's Hybrane hyperbranched polyesteramides. *Journal of Polymer Science Part A: Polymer Chemistry* **42,** 3110–3115. doi:10.1002/pola.20113 (2004).

2. Voit, B. I. & Lederer, A. Hyperbranched and Highly Branched Polymer Architectures—Synthetic Strategies and Major Characterization Aspects. *Chemical Reviews* **109,** 5924–5973. doi:10.1021/cr900068q (2009).

3. Prado-Socorro, C. D., Giménez-Santamarina, S., Mardegan, L., *et al.* Polymer-Based Composites for Engineering Organic Memristive Devices. *Advanced Electronic Materials* **8,** 2101192. doi:10.1002/aelm.202101192 (2022).